

# Deep learning for longevity projections. State of art and future directions

EDSD workshop on Machine Learning

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# Outline

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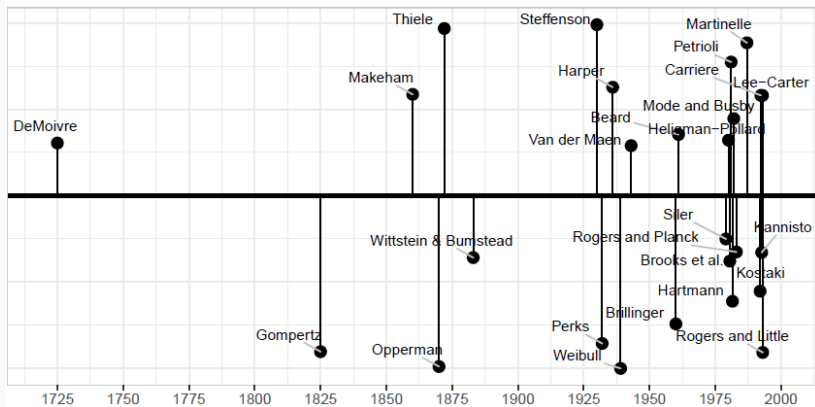
# Outline

- What's deep learning
- Deep Learning for mortality modeling
- Discussion on Mortality Modeling
- Conclusions

# Intro Mortality modeling

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# Mortality Modelling. Crash Course



# Model: Lee-Carter

LC proposes a log-bilinear model for mortality rates, incorporating both age and year effects:

$$\ln(m_{ij}) = \alpha_i + \beta_i \kappa_j + \epsilon_{ij} \quad (1)$$

$\alpha_i$  is the general shape of the log-mortality at age  $i$

$\kappa_j$  represents the time trend index of general mortality level

$\beta_i$  indicates the sensitivity of the log-mortality at age  $i$  to variations in the time index

$\epsilon_{ij}$  is the residual term at age  $x$  and time  $t$ .

With constraints:

$$\sum_i \beta_i^2 = 1; \sum_j \kappa_j = 0 \quad (2)$$

# Mortality Modelling. Crash Course

Lee-Carter model has been used and revised since 1992

- Booth and Tickle (2008), Booth et al. (2006), Li and Lee (2005), Renshaw and Haberman ... Nigri et al. (2019)
- its canonical version is still broadly used as a benchmark.  
The scientific environment has been dominated by Lee Carter model approach

# Mortality Modelling. Crash Course

## Parallel Approaches

- Life Expectancy Forecast:
  - Oeppen and Vaupel (2002)
  - Torri and Vaupel (2012)
  - Lee (2006)
  - Raftery et al. (2013)
  - Pascariu et al. (2018)
  - Nigri et al. (2021)
- Age at Death distribution
  - Basellini and Camarda (2019)
  - Pascariu et al. (2018)



# Deep Learning for mortality modeling

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# Deep Learning for mortality modeling

- Integration
- Forecasting: Death rates and summary demographic measures
- Indirect estimation

# DL as Integration of mortality models



Article

## A Deep Learning Integrated Lee-Carter Model

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**Abstract:** In the field of mortality forecasting, the Lee-Carter model is a benchmark for comparing the performance of different models. The traditional ARIMA process mortality shape. Concerning order to think a nonlinear approach the ARIMA process the pattern of  $x_t$  series over Short-Term Memory archite. The proposed approach proposes for avoiding the time-churn delete the time in which the Long Short-Term Memory into the forecasted trend, due patterns.

**Keywords:** mortality; deep learning

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## A Neural Approach to Improve the Lee-Carter Mortality Density Forecasts

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Several countries worldwide are experiencing a continuous increase in life expectancy, extending the challenges of life actuaries and demographers in forecasting mortality. Although several stochastic mortality models have been proposed in the literature, mortality forecasting research remains a crucial task. Recently, various research works have encouraged the use of deep learning models to extrapolate suitable patterns within mortality data. Such learning models allow achieving accurate point predictions, though uncertainty measures are also necessary to support both model estimate reliability and risk evaluation. As a new advance in mortality forecasting, we formalize the deep neural network integration within the Lee-Carter framework, as a first bridge between the deep learning and the mortality density forecasts. We test our model proposal in a numerical application considering three representative countries worldwide and for both genders, scrutinizing two different fitting periods. Exploiting the meaning of both biological reasonableness and plausibility of forecasts, as well as performance metrics, our findings confirm the suitability of deep learning models to improve the predictive capacity of the Lee-Carter model, providing more reliable mortality boundaries in the long run.

# DL as Integration of mortality models

A DL Integrated Lee-Carter model

- Too long fitting period, age structure issues

A Neural Approach to Improve the Lee-Carter Mortality Density Forecast

- Improvements: Shorter fitting period, prediction intervals estimation

TABLE 2  
log  $m_{x,t}$  Performance Metrics Values for each Training Period, Forecasting Years: 2001–2018

|           |         | x = 45                    |              |              |              |              |              |                           |              |              |              |              |              |
|-----------|---------|---------------------------|--------------|--------------|--------------|--------------|--------------|---------------------------|--------------|--------------|--------------|--------------|--------------|
| Country   | Model   | Training period 1950-2000 |              |              |              |              |              | Training period 1960-2000 |              |              |              |              |              |
|           |         | Male                      |              |              | Female       |              |              | Male                      |              |              | Female       |              |              |
|           |         | $RMSE_{(m)}$              | $PICP_{(m)}$ | $MPIW_{(m)}$ | $RMSE_{(m)}$ | $PICP_{(m)}$ | $MPIW_{(m)}$ | $RMSE_{(m)}$              | $PICP_{(m)}$ | $MPIW_{(m)}$ | $RMSE_{(m)}$ | $PICP_{(m)}$ | $MPIW_{(m)}$ |
| Australia | LC      | 0.227                     | <b>1</b>     | <b>0.534</b> | <b>0.091</b> | 0.944        | 0.267        | 0.175                     | <b>1</b>     | <b>0.478</b> | <b>0.084</b> | 0.944        | 0.265        |
|           | LC-LSTM | <b>0.110</b>              | 0.944        | 0.295        | 0.142        | 0.944        | <b>0.407</b> | <b>0.116</b>              | 0.944        | 0.280        | 0.097        | <b>1</b>     | <b>0.394</b> |
| Japan     | LC      | 0.071                     | 0.667        | <b>0.180</b> | 0.255        | 0            | 0.173        | <b>0.063</b>              | 0.722        | 0.150        | 0.155        | 0.056        | 0.105        |
|           | LC-LSTM | <b>0.062</b>              | <b>0.722</b> | 0.143        | <b>0.077</b> | <b>0.444</b> | <b>0.254</b> | 0.073                     | 0.944        | 0.243        | <b>0.061</b> | <b>0.667</b> | <b>0.115</b> |
| Spain     | LC      | 0.200                     | 0.333        | 0.153        | <b>0.104</b> | 0.611        | 0.179        | 0.228                     | <b>0.333</b> | 0.136        | <b>0.067</b> | 0.722        | 0.174        |
|           | LC-LSTM | <b>0.161</b>              | <b>0.556</b> | <b>0.276</b> | 0.502        | <b>0.944</b> | <b>0.489</b> | <b>0.205</b>              | 0.278        | <b>0.215</b> | 0.073        | <b>0.944</b> | <b>0.259</b> |
|           |         | x = 65                    |              |              |              |              |              |                           |              |              |              |              |              |
| Country   | Model   | Training period 1950-2000 |              |              |              |              |              | Training period 1960-2000 |              |              |              |              |              |
|           |         | Male                      |              |              | Female       |              |              | Male                      |              |              | Female       |              |              |
|           |         | $RMSE_{(m)}$              | $PICP_{(m)}$ | $MPIW_{(m)}$ | $RMSE_{(m)}$ | $PICP_{(m)}$ | $MPIW_{(m)}$ | $RMSE_{(m)}$              | $PICP_{(m)}$ | $MPIW_{(m)}$ | $RMSE_{(m)}$ | $PICP_{(m)}$ | $MPIW_{(m)}$ |
| Australia | LC      | 0.157                     | <b>1</b>     | <b>0.672</b> | 0.061        | 0.944        | 0.283        | 0.106                     | <b>1</b>     | 0.623        | 0.058        | <b>1</b>     | 0.293        |
|           | LC-LSTM | <b>0.056</b>              | <b>1</b>     | 0.371        | 0.061        | <b>1</b>     | <b>0.431</b> | 0.106                     | <b>1</b>     | 0.365        | 0.052        | <b>1</b>     | 0.436        |
| Japan     | LC      | 0.054                     | <b>1</b>     | <b>0.177</b> | 0.160        | 0.444        | 0.178        | 0.063                     | 0.833        | 0.161        | 0.151        | 0.333        | 0.128        |
|           | LC-LSTM | <b>0.035</b>              | 0.944        | 0.141        | 0.077        | <b>1</b>     | 0.262        | 0.029                     | <b>1</b>     | 0.261        | 0.028        | <b>1</b>     | 0.141        |
| Spain     | LC      | 0.097                     | 0.278        | 0.157        | 0.079        | 0.778        | 0.206        | 0.106                     | 0.222        | 0.158        | 0.073        | 0.889        | 0.229        |
|           | LC-LSTM | <b>0.060</b>              | <b>1</b>     | <b>0.285</b> | 0.066        | <b>1</b>     | 0.568        | 0.080                     | 0.889        | 0.249        | 0.068        | 0.944        | 0.340        |
|           |         | x = 85                    |              |              |              |              |              |                           |              |              |              |              |              |
| Country   | Model   | Training period 1950-2000 |              |              |              |              |              | Training period 1960-2000 |              |              |              |              |              |
|           |         | Male                      |              |              | Female       |              |              | Male                      |              |              | Female       |              |              |
|           |         | $RMSE_{(m)}$              | $PICP_{(m)}$ | $MPIW_{(m)}$ | $RMSE_{(m)}$ | $PICP_{(m)}$ | $MPIW_{(m)}$ | $RMSE_{(m)}$              | $PICP_{(m)}$ | $MPIW_{(m)}$ | $RMSE_{(m)}$ | $PICP_{(m)}$ | $MPIW_{(m)}$ |
| Australia | LC      | 0.053                     | 0.944        | 0.344        | 0.032        | <b>1</b>     | 0.191        | 0.039                     | 0.944        | 0.319        | 0.033        | <b>1</b>     | 0.194        |
|           | LC-LSTM | 0.056                     | 0.944        | 0.190        | 0.033        | <b>1</b>     | 0.292        | 0.049                     | 0.944        | 0.187        | 0.026        | <b>1</b>     | 0.289        |
| Japan     | LC      | 0.030                     | 0.889        | 0.134        | 0.050        | 0.778        | 0.142        | 0.040                     | 0.944        | 0.133        | 0.071        | 0.444        | 0.115        |
|           | LC-LSTM | 0.034                     | 0.778        | 0.107        | 0.171        | 0.500        | 0.209        | 0.029                     | 0.944        | 0.215        | 0.080        | 0.444        | 0.126        |
| Spain     | LC      | 0.082                     | 0.333        | 0.113        | 0.059        | 0.611        | 0.122        | 0.086                     | 0.278        | 0.116        | 0.057        | 0.833        | 0.150        |
|           | LC-LSTM | 0.052                     | <b>1</b>     | 0.204        | 0.447        | <b>1</b>     | 0.335        | 0.066                     | 0.944        | 0.183        | 0.048        | <b>1</b>     | 0.223        |

# Forecasting: Death rates

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PAPER



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and Faculty  
of Actuaries

## A neural network extension of the Lee–Carter model to multiple populations

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(Received 31 October



Abstract  
The Lee–Cart

## CALIBRATING THE LEE-CARTER AND THE POISSON LEE-CARTER MODELS VIA NEURAL NETWORKS

Published online by Cambridge University Press: 31 March 2022

Salvatore Scognamiglio

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## POINT AND INTERVAL FORECASTS OF DEATH RATES USING NEURAL NETWORKS

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Simon Schnürch and Ralf Korn

SCANDINAVIAN ACTUARIAL JOURNAL  
<https://doi.org/10.1080/03461238.2020.1867232>



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## Time-series forecasting of mortality rates using deep learning

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### ABSTRACT

The time-series nature of mortality rates lends itself to processing through neural networks that are specialized to deal with sequential data, such as recurrent and convolutional networks. The aim of this work is to show how the structure of the Lee–Carter model can be generalized using a relatively simple shallow convolutional network model, allowing for its components to be evaluated in familiar terms. Although deep networks have been applied successfully in many areas, we find that deep networks do not lead to an enhanced predictive performance in our approach for mortality forecasting, compared to the proposed shallow one. Our model produces highly accurate forecasts on the Human Mortality Database, and,

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### KEYWORDS

Mortality forecasting;  
recurrent neural networks;  
convolutional neural  
networks; representation  
learning; time-series  
forecasting; Lee–Carter  
model; human mortality



ASTIN Bulletin: The  
Journal of the IAA

### Article contents

Abstract

Introduction

Neural network  
models

Prediction  
uncertainty

Article Figures Supplementary material

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### Abstract

The Lee–Carter model has become a forecasting performance can be significant techniques. We propose a convolutional forecasting, empirically compare this model and find that lower forecast error Mortality Database. We provide detail more understandable and, thus, more

# Forecasting: Death rates

## Cons:

- Training /Test approach ... no autoregressive framework.
- Single fitting period.

## Pros:

- Make multi-population dimension easy.

TABLE 4.

MSE, MAE AND MAPE OF LC\_SVD, LC\_POISSON, RH AND LC\_LCN\_POISSON IN THE AGE RANGES 0–99 AND 55–90 FOR EACH GENDER; MSE AND MAE VALUES ARE IN  $10^{-4}$  ON THE 69 POPULATIONS IN WHICH THE RH MODEL CONVERGES; THE BEST PERFORMANCE IS REPORTED IN BOLD.

| Age range | Model          | MSE           |               | MAE            |                | MAPE           |                |
|-----------|----------------|---------------|---------------|----------------|----------------|----------------|----------------|
|           |                | Female        | Male          | Female         | Male           | Female (%)     | Male (%)       |
| 0–99      | LC_SVD         | 2.6103        | 5.1177        | 55.0237        | 94.9656        | 25.3196        | 35.8088        |
|           | LC_Poisson     | 2.3962        | 4.7243        | 49.7804        | 81.8419        | 25.4649        | 34.6300        |
|           | RH             | 41.3951       | 6.7771        | 95.9712        | 89.7170        | 27.4428        | <b>32.0161</b> |
|           | LC_LCN_Poisson | <b>1.7988</b> | <b>2.8346</b> | <b>42.4132</b> | <b>60.9455</b> | <b>23.5324</b> | 35.9236        |
| 55–90     | LC_SVD         | 1.1918        | 3.3327        | 61.7312        | 119.9267       | 17.7076        | 25.3430        |
|           | LC_Poisson     | 1.1272        | 2.8842        | 53.5945        | 110.0120       | 15.8867        | 22.0608        |
|           | RH             | 3.0516        | 4.1034        | 65.8552        | 107.2838       | 16.1086        | 16.8226        |
|           | LC_LCN_Poisson | <b>0.4682</b> | <b>1.3546</b> | <b>40.7667</b> | <b>76.9406</b> | <b>12.1228</b> | <b>15.4465</b> |



# Forecasting: Summary demographic measures

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## Clustering-based simultaneous forecasting of life expectancy time series through Long-Short Term Memory Neural Networks



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## Life expectancy and lifespan disparity forecasting: a long short-term memory approach

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### ABSTRACT

After the World War II, developed countries experienced a constant decline in mortality. As a result, life expectancy has never stopped increasing, despite an evident deceleration in developed countries, e.g. England, USA and Denmark. In this paper, we propose a new approach for forecasting life expectancy and lifespan disparity based on the recurrent neural networks with a long short-term memory. This type of neural network leads to predicting future values of longevity, while maintaining the significant

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### KEYWORDS

Life expectancy; lifespan disparity; forecasting; long short-term memory

**Table 1.** Out-of-sample test for  $e_{0,t}$ : MAE and RMSE for LSTM, ARIMA, DG, LC and CoDa model by country and gender.

| Country   | Model | Fitting period: 1938–1999 |               |               |               | Fitting period: 1947–1999 |               |               |               |
|-----------|-------|---------------------------|---------------|---------------|---------------|---------------------------|---------------|---------------|---------------|
|           |       | Female                    |               | Male          |               | Female                    |               | Male          |               |
|           |       | MAE                       | RMSE          | MAE           | RMSE          | MAE                       | RMSE          | MAE           | RMSE          |
| Australia | ARIMA | 0.3118                    | 0.4149        | 0.8111        | 0.8504        | 0.2167                    | 0.2844        | 0.2901        | 0.3216        |
|           | DG    | 0.3139                    | 0.4175        | 0.2693        | 0.2896        | 0.1945                    | 0.2277        | 0.3219        | 0.3468        |
|           | LSTM  | <b>0.1139</b>             | <b>0.1412</b> | <b>0.1485</b> | <b>0.1895</b> | <b>0.1110</b>             | <b>0.1362</b> | <b>0.1407</b> | <b>0.1804</b> |
|           | LC    | 0.2525                    | 0.2806        | 1.0740        | 1.2133        | 0.2869                    | 0.3204        | 1.0368        | 1.1640        |
|           | CoDa  | 0.1347                    | 0.1655        | 1.1936        | 1.2763        | 0.1304                    | 0.1639        | 1.0022        | 1.0629        |
| Italy     | ARIMA | 1.5759                    | 1.8872        | 0.9157        | 1.0819        | 0.3434                    | 0.4455        | 0.1768        | 0.2155        |
|           | DG    | 0.2986                    | 0.3836        | 0.2355        | 0.2697        | 0.2314                    | 0.2722        | 0.2209        | 0.2444        |
|           | LSTM  | 0.1914                    | 0.2304        | <b>0.1396</b> | <b>0.1767</b> | 0.2104                    | 0.2587        | <b>0.1758</b> | <b>0.2124</b> |
|           | LC    | <b>0.1663</b>             | <b>0.2068</b> | 1.8194        | 1.9259        | <b>0.1518</b>             | <b>0.1969</b> | 1.5136        | 1.6463        |
|           | CoDa  | 0.4275                    | 0.5507        | 0.9763        | 1.0531        | 0.4156                    | 0.5356        | 1.0985        | 1.1880        |
| Sweden    | ARIMA | 0.4305                    | 0.4659        | 0.4760        | 0.5484        | 0.4467                    | 0.4672        | 0.2696        | 0.3058        |
|           | DG    | 0.4305                    | 0.4659        | 0.1659        | 0.1888        | 0.4467                    | 0.4671        | 0.3983        | 0.4232        |
|           | LSTM  | <b>0.0773</b>             | <b>0.0964</b> | <b>0.0574</b> | <b>0.0703</b> | <b>0.0752</b>             | <b>0.1000</b> | <b>0.0598</b> | <b>0.0718</b> |
|           | LC    | 0.1761                    | 0.1973        | 0.9698        | 1.0815        | 0.0823                    | 0.1149        | 1.0199        | 1.1245        |
|           | CoDa  | 0.4079                    | 0.4574        | 0.9496        | 1.0627        | 0.6612                    | 0.7449        | 0.8578        | 0.9571        |
| USA       | ARIMA | 0.7358                    | 0.8898        | 0.1892        | 0.2449        | 0.6822                    | 0.8165        | 0.1455        | 0.1845        |
|           | DG    | 0.7358                    | 0.8898        | 1.3553        | 1.5444        | 0.6821                    | 0.8164        | 1.0669        | 1.2119        |
|           | LSTM  | 0.2466                    | 0.2939        | <b>0.1140</b> | <b>0.1381</b> | 0.3522                    | 0.4279        | <b>0.1137</b> | <b>0.1375</b> |
|           | LC    | <b>0.1173</b>             | <b>0.1451</b> | 0.5017        | 0.5950        | 0.3847                    | 0.4096        | 0.7549        | 0.8336        |
|           | CoDa  | 0.2390                    | 0.2688        | 0.4505        | 0.5432        | <b>0.1038</b>             | <b>0.1266</b> | 0.4529        | 0.5451        |
| Japan     | ARIMA | –                         | –             | –             | –             | <b>0.1712</b>             | <b>0.2291</b> | 1.2220        | 1.4085        |
|           | DG    | –                         | –             | –             | –             | 0.5569                    | 0.5894        | 0.3721        | 0.4210        |
|           | LSTM  | –                         | –             | –             | –             | 0.3342                    | 0.3694        | <b>0.2252</b> | <b>0.2662</b> |
|           | LC    | –                         | –             | –             | –             | 0.6543                    | 0.7650        | 0.4330        | 0.5068        |
|           | CoDa  | –                         | –             | –             | –             | 1.2086                    | 1.5032        | 0.9961        | 1.2106        |

Notes: Years 2000–2014. Fitting periods: 1938–1999 (columns 3–6) and 1947–1999 (columns 7–10).

**Table 3.** Out-of-sample test for  $e_{0,t}^{\dagger}$ : MAE and RMSE for LSTM, ARIMA, LC and CoDa model by country and gender.

| Country   | Model | Fitting period: 1938–1999 |               |               |               | Fitting period: 1947 – 1999 |               |               |               |
|-----------|-------|---------------------------|---------------|---------------|---------------|-----------------------------|---------------|---------------|---------------|
|           |       | Female                    |               | Male          |               | Female                      |               | Male          |               |
|           |       | MAE                       | RMSE          | MAE           | RMSE          | MAE                         | RMSE          | MAE           | RMSE          |
| Australia | ARIMA | 0.0916                    | 0.0985        | 0.1058        | 0.1348        | <b>0.0718</b>               | 0.0994        | 0.1426        | 0.1618        |
|           | LSTM  | 0.0906                    | 0.1018        | <b>0.0631</b> | <b>0.0756</b> | 0.0794                      | 0.0929        | 0.0880        | 0.1103        |
|           | LC    | <b>0.0729</b>             | 0.0949        | 0.0844        | 0.1150        | 0.1931                      | 0.2041        | <b>0.0764</b> | <b>0.0981</b> |
|           | CoDa  | 0.0757                    | <b>0.0864</b> | 0.2147        | 0.2270        | 0.0845                      | <b>0.0924</b> | 0.1601        | 0.1785        |
| Italy     | ARIMA | 0.3209                    | 0.3709        | 0.9104        | 1.0565        | 0.5013                      | 0.5810        | 0.3444        | 0.3955        |
|           | LSTM  | <b>0.0545</b>             | <b>0.0643</b> | <b>0.0702</b> | <b>0.0866</b> | <b>0.1362</b>               | <b>0.1528</b> | <b>0.0646</b> | <b>0.0827</b> |
|           | LC    | 0.2100                    | 0.2222        | 0.3984        | 0.4641        | 0.1649                      | 0.1792        | 0.3860        | 0.4513        |
|           | CoDa  | 0.2451                    | 0.2833        | 0.2807        | 0.3247        | 0.2331                      | 0.2702        | 0.3533        | 0.4073        |
| Sweden    | ARIMA | 0.2438                    | 0.2666        | 0.3020        | 0.3390        | 0.2944                      | 0.3262        | 0.2166        | 0.2442        |
|           | LSTM  | <b>0.0598</b>             | <b>0.0736</b> | <b>0.0468</b> | <b>0.0550</b> | <b>0.0572</b>               | <b>0.0669</b> | <b>0.0439</b> | <b>0.0565</b> |
|           | LC    | 0.1204                    | 0.1291        | 0.0559        | 0.0734        | 0.1025                      | 0.1126        | 0.1798        | 0.1955        |
|           | CoDa  | 0.2195                    | 0.2398        | 0.0612        | 0.0729        | 0.2379                      | 0.2634        | 0.0771        | 0.0920        |
| USA       | ARIMA | 0.5569                    | 0.6499        | 0.8677        | 0.9733        | 0.4795                      | 0.5508        | 0.5935        | 0.6626        |
|           | LSTM  | <b>0.0457</b>             | <b>0.0547</b> | <b>0.0497</b> | <b>0.0603</b> | <b>0.0517</b>               | <b>0.0561</b> | <b>0.0529</b> | <b>0.0628</b> |
|           | LC    | 0.1670                    | 0.2006        | 0.3277        | 0.3742        | 0.3281                      | 0.3514        | 0.1659        | 0.1931        |
|           | CoDa  | 0.3269                    | 0.3885        | 0.4529        | 0.5027        | 0.1970                      | 0.2433        | 0.4246        | 0.4715        |
| Japan     | ARIMA | –                         | –             | –             | –             | 0.0635                      | 0.0868        | 0.2265        | 0.2863        |
|           | LSTM  | –                         | –             | –             | –             | <b>0.0573</b>               | <b>0.0760</b> | <b>0.0726</b> | <b>0.0799</b> |
|           | LC    | –                         | –             | –             | –             | 1.1321                      | 1.1351        | 0.7256        | 0.7276        |
|           | CoDa  | –                         | –             | –             | –             | 0.9129                      | 0.9958        | 0.5617        | 0.6017        |

Notes: Years 2000–2014. Fitting periods: 1938–1999 (columns 3–6) and 1947–1999 (columns 7–10).

# Indirect Estimation

- Demography has a long-standing tradition in indirect estimation approach (United Nations, 1955, 1967; Murray et al., 2003, Brass et al., 1968;1971,Wilmoth et al., 2012, Ševčíková et al. 2016 and Pascariu et al. 2020)
- **"Indirect"** qualifies the demographic estimation technique that produces estimates of parameters using information only indirectly related to its value.



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*Research Article*

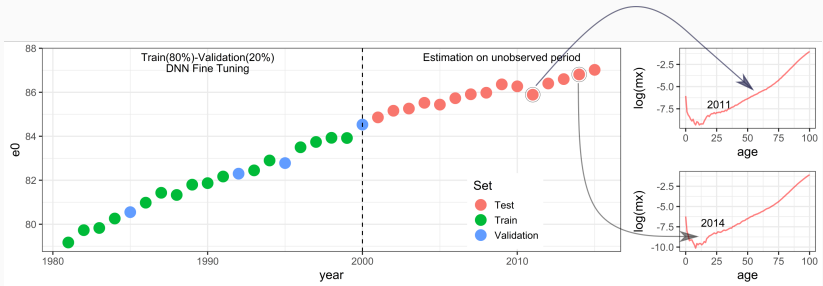
**Leveraging deep neural networks to estimate  
age specific mortality from life expectancy at  
birth**

**Andrea Nigri**

**Susanna Levantesi**

**José Manuel Aburto**

# Indirect Estimation



# Indirect Estimation: DNN model

- Contribution to the literature on the demographic indirect estimation using a Deep Neural Network framework.
- The model could be used to indirectly estimate vital rates for regions or subpopulations with similar mortality profiles.
- This characteristic makes the proposed model appealing for countries where present information is lacking but past data are available or from surrounding countries or populations.

# Discussion on Mortality Modeling

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# Discussion on Mortality Modeling

What are the most recent important models?

# Discussion on Mortality Modeling

Attempt to answer the question:

- Pascariu MD, Lenart A, Canudas-Romo V. (2019) The maximum entropy mortality model: forecasting mortality using statistical moments. Scandinavian Actuarial Journal 2019 (8), 661-685

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- Aliverti, E., Mazzuco, S. and Scarpa B. (2022) Dynamic modeling of mortality via mixtures of skewed distribution functions. *Journal of the Royal Statistical Society, Series A*

# Despite a constant proposal of innovative contributions...



# Checklist of criteria - Cairns et al (2011)

- Long-term dynamics under the model should be biologically reasonable,
- The robustness of predictions w.r.t. the ages and historical mortality trend
- The model should be straightforward to implement using analytical methods or fast numerical algorithms.
- Also...

The plausibility of projected uncertainty at different ages. We can use performance metric to assess the mortality forecasts with a back-testing approach.

If you want to know more about Lee-Carter: "*Thirty years on: A review of the Lee-Carter method for forecasting mortality.*" U Basellini, CG. Camarda, and H. Booth. (Preprint on SocArxiv)

**30th and 20th anniversary** of two of the most important works in the study of longevity:

- Lee, Ronald D., and Lawrence R. Carter. "Modeling and Forecasting U. S. Mortality." Journal of the American Statistical Association. 1992
- Oeppen J, Vaupel JW. Demography. Broken limits to life expectancy. Science. 2002

# Food for thought

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# Test your model

- Test your model on summary measures (Bohk-Ewald et al. (2017))

|             | Lee-Carter | R&H    | CBD    | ...    | CoDa   |
|-------------|------------|--------|--------|--------|--------|
| $e_0$       | (RMSE)     |        |        |        |        |
| $e_{65}$    |            | (RMSE) |        |        |        |
| $\uparrow$  |            |        |        |        |        |
| $e_0$       |            |        | (RMSE) |        |        |
| $\uparrow$  |            |        |        |        |        |
| $e_{65}$    |            |        |        | (RMSE) |        |
| $Gini_0$    |            |        |        |        | (RMSE) |
| $Gini_{65}$ |            |        |        |        |        |

- Leverage forecasting innovations in other areas ...  
Fertility? Register-based population study?

# Literature evolution



# Conclusions

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# Conclusions

- Forecasting future longevity is not straightforward
- The accuracy of population projections should be regularly tested to ensure evidence of improvement.
- Deep Learning in Longevity: Actuarial journals Vs. Demographic journals...
- Considering a new and different perspective, can help the community to better understand how (and if) the field can really benefit from this new approach.

**Thank you for your  
attention.**

# References

- S. Levantesi, A. Nigri, and G. Piscopo. Clustering-based simultaneous forecasting of life expectancy time series through long-short term memory neural networks. *International Journal of Approximate Reasoning*, 140:282–297, 2022.
- M. Marino, S. Levantesi, and A. Nigri. A neural approach to improve the lee-carter mortality density forecasts. *North American Actuarial Journal*, pages 1–18, 2022.
- A. Miyata and N. Matsuyama. Extending the lee-carter model with variational autoencoder: A fusion of neural network and bayesian approach. *ASTIN Bulletin: The Journal of the IAA*, pages 1–24.
- A. Nigri, S. Levantesi, and J. Aburto. Leveraging deep neural networks to estimate age-specific mortality from life expectancy at birth. *Demographic Research*, 47, 2022.
- A. Nigri, S. Levantesi, and M. Marino. Life expectancy and lifespan disparity forecasting: a long short-term memory approach. *Scandinavian Actuarial Journal*, 2021(2):110–133, 2021.
- A. Nigri, S. Levantesi, M. Marino, S. Scognamiglio, and F. Perla. A deep learning integrated lee-carter model. *Risks*, 7(1):33, 2019.
- F. Perla, R. Richman, S. Scognamiglio, and M. V. Wuthrich. Time-series forecasting of mortality rates using deep learning. *Scandinavian Actuarial Journal*, pages 1–27, 2021.
- R. Richman. Ai in actuarial science—a review of recent advances—part 1. *Annals of Actuarial Science*, 15(2):207–229, 2021.
- R. Richman. Ai in actuarial science—a review of recent advances—part 2. *Annals of Actuarial Science*, 15(2):230–258, 2021.
- R. Richman and M. V. Wuthrich. Lee and carter go machine learning: recurrent neural networks. Available at SSRN 3441030, 2019.
- R. Richman and M. V. Wuthrich. A neural network extension of the lee-carter model to multiple populations. *Annals of Actuarial Science*, 15(2):346–366, 2021.
- S. Schnurch and R. Korn. Point and interval forecasts of death rates using neural networks. *ASTIN Bulletin: The Journal of the IAA*, 52(1):333–360, 2022.
- S. Scognamiglio. Calibrating the lee-carter and the poisson lee-carter models via neural networks. *ASTIN Bulletin: The Journal of the IAA*, pages 1–43, 2022.